

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456719.78724 +/- 0.00098 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.066 +/- 0.047
Transit depth (Rp/Rs)^2	0.44 +/- 0.63 [%]
Orbital Inclination (inc)	79.67 +/- 3.03
Airmass coefficient 1 (a1)	3.67 +/- 1.94
Airmass coefficient 2 (a2)	-0.85 +/- 0.43
Scatter in the residuals of the lightcurve fit is	56.69 %
Best Comparison Star	None
Optimal Aperture	2.05
Optimal Annulus	7.5
Transit Duration (day)	0.029 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.066 $\pm$ 0.047 vs 0.1368 (deviation 1.27 σ)
Duration fit vs published	0.029 d vs 0.0758 d (deviation 1.32 σ)
Mid-time O-C	-0.2 min
Depth SNR	1.2
Residual scatter	56.69 %

Light curve

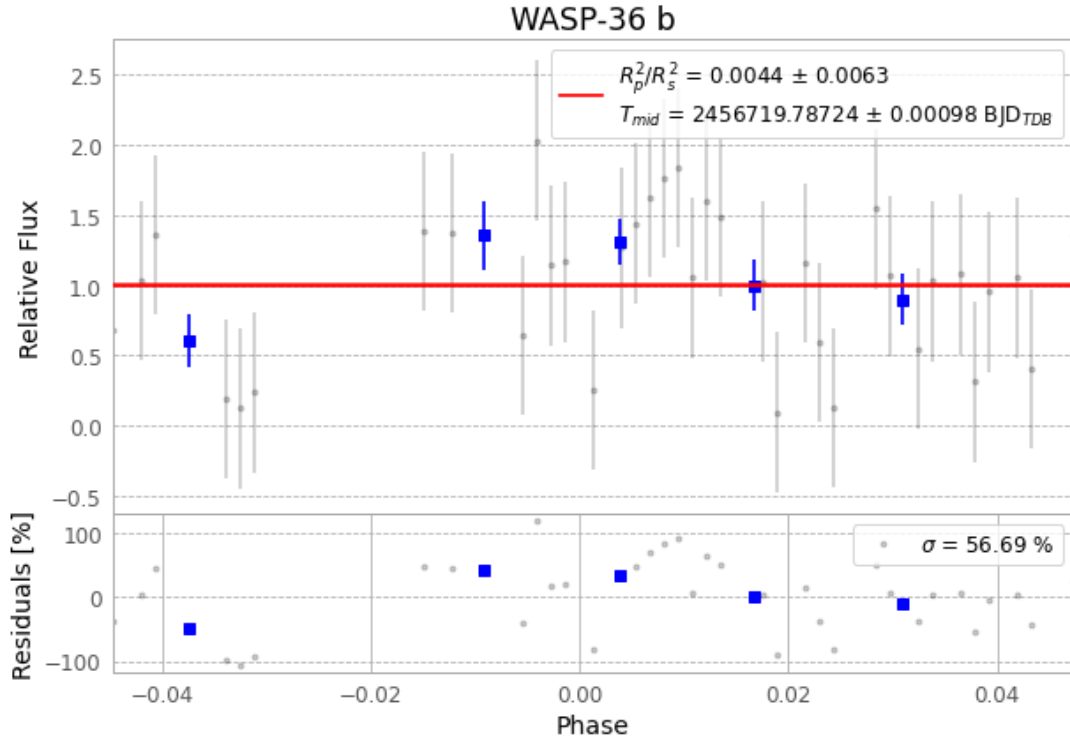


Figure 1 — FinalLightCurve_WASP-36 b_2014-03-02.png (SHA-256 35d561c775cf470f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [492, 458], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9f9cd8fc2719235d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

68 FITS frames (45 MB), WASP-36140303050612.FITS → WASP-36140303083012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-140303.log (928f00b35e95...), FinalLightCurve_WASP-36 b_2014-03-02.png (35d561c775cf...), FinalLightCurve_WASP-36 b_2014-03-02.csv (07e8861847fb...)

Compute resources

Wall time 188.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 01:15Z.